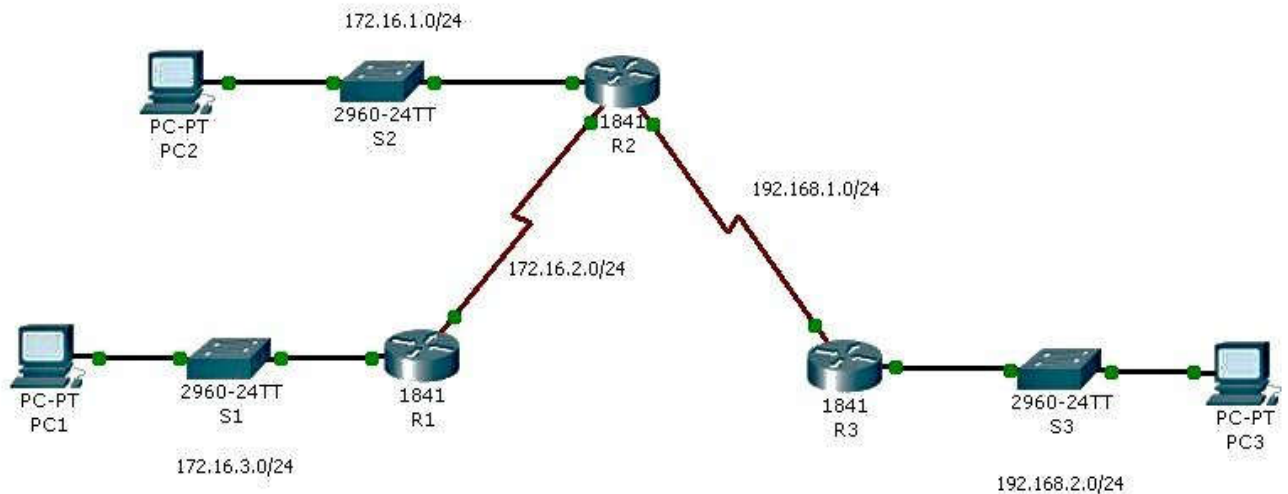


The Computer Networks

Lab 5

TP Static Routing - 3

The purpose of this Lab is to implement a static routing configuration.



5. Verification of configuring a static route

Presentation:

During this exercise, we will study the topology configured with static routes, which refer to the IP address of next hop and require a recursive search. We will replace these static routes by others that specify the outgoing interfaces and which require no recursive search. We will also examine the effect of such a change.

Educational goals:

- Examine the network with static routes of the next hop
- Check the configuration
- Check connectivity
- Update the network using static routes to the outgoing interface
- Delete static routes of the next hop
- Configure static routes to the outgoing interface

- Examine the network with static routes to the outgoing interface
- Check the configuration
- Check connectivity

Task 1: Network examination with static routes next hop

Step 1: Check the configuration.

Proceed as follows for the three routers:

Enter the command **show running-config** to determine the current configuration of static routing.

Enter the command **show ip route** to see the effect of this configuration. Each static route must perform a recursive search on a connected route to forward a packet.

Step 2: Verify connectivity.

On the three computers, perform a ping test from the command line prompt of each one to the other two computers. All ping tests should not succeed.

Task 2: Network update using static routes to the outgoing interface

Step 1 - Set up static routes to the outgoing interface.

Enter the following command on all three routers:

R1

```
R1(config)#ip route 172.16.1.0 255.255.255.0 s0/0/0
```

```
R1(config)#ip route 192.168.1.0 255.255.255.0 s0/0/0
```

```
R1(config)#ip route 192.168.2.0 255.255.255.0 s0/0/0
```

R2

```
R2(config)#ip route 172.16.3.0 255.255.255.0 s0/0/0
```

```
R2(config)#ip route 192.168.2.0 255.255.255.0 s0/0/1
```

R3

```
R3(config)#ip route 172.16.1.0 255.255.255.0 s0/0/1
```

```
R3(config)#ip route 172.16.2.0 255.255.255.0 s0/0/1
```

```
R3(config)#ip route 172.16.3.0 255.255.255.0 s0/0/1
```

Step 2: Save the updated configurations.

Exit configuration mode on all three routers using the keyboard shortcut Ctrl + Z. Save the configuration by running the command **copy run start**.

Step 3: Delete routes

Switch to Global Configuration mode on the router from which you want to delete an entry from its routing table and enter the following command:

For router R1, for example:

```
R1(config)#no ip route 172.16.1.0 255.255.255.0 172.16.2.2
```

Attention, it is necessary to choose an existing entry to remove it. If you do it for the test do not forget to put the static route to continue the Lab without problem.

Task 3: Network exam with static routes to the outgoing interface

Step 1: Check the configuration.

Follow these steps on all three routers:

Enter the command **show running-config** to view the changed configuration.

Enter the command **show ip route** to examine the effect of the changed configuration. Each static route is connected directly and no recursive search is required to transfer a packet.

Step 2: Verify connectivity.

Ping the other two computers in the command-line prompt on all three computers. All ping tests must succeed. If not, troubleshoot static routes.